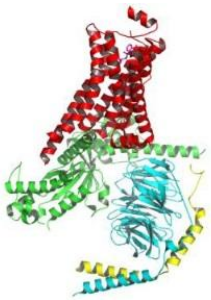
One Pot Aldol Condensation and Diels-Alder Experiment
X-ray Crystal Structure Data

The technique of X-ray crystallography is one of the most important analytical tools developed for understanding the arrangement of atoms within molecules, small and large. Its importance is reflected in the 26 Nobel Prizes awarded which were directly related to or involved the use of crystallographic methods. For example:

1915 Physics W.H. Bragg and W.L. Bragg *Use of X-rays to determine crystal structure*
1936 Chemistry Petrus (Peter) Josephus Wilhelmus Debye *For his contributions to our knowledge of molecular structure through his investigations on dipole moments and on the diffraction of X-rays and electrons in gases*
1954 Chemistry L.C. Pauling *For his research into the nature of the chemical bond and its application to the elucidation of the structure of complex substances*
2009 Chemistry V. Ramakrishnan, T.A. Steitz, A.E. Yonath *Studies of the structure and function of the ribosome*
2011 Chemistry D. Shechtman *For the discovery of quasicrystals*

In addition, a wide range of scientific disciplines (chemistry, biology, geology, physics and even math) have relied on X-ray crystallography for new discoveries and advancing their technologies. As evidence of its importance the United Nations Educational, Scientific, and Cultural Organization has designated 2014 as the "International Year of Crystallography." Go to http://www.iycr2014.org/ and http://www.youtube.com/watch?v=RkRYe1J7cHE for an interesting overview of the field.

Perhaps one of the most extensive and impressive applications of X-ray crystallography has been for protein and nucleic acid structures. In, 2012 the Nobel Prize in Chemistry was awarded to Robert J. Lefkowitz and Brain K. Kobika for studies of G-protein-coupled receptors which are responsible for cell membrane signaling.[1] X-ray crystal structure data collected for G-proteins bound to agonists was key to understanding their biochemical mechanisms and aided in the development of important drugs. Go to the Protein Data Bank by following this link http://www.rcsb.org/pdb/home/home.do and enter the pdb ID: 2rh1. Go to the far right where a 3-dimensional biomolecule is shown and click “View in 3D”. Download software if needed to view the Jmol structure from various perspectives.



[1] "The Nobel Prize in Chemistry 2012". Nobelprize.org. 2 Nov 2012 http://www.nobelprize.org/nobel_prizes/chemistry/laureates/2012/

Though the instrumentation and theory of X-ray crystallography is well beyond the scope of this course, some key concepts as they apply to the small molecules we are preparing in the lab will be summarized here. First, the method of crystallography described and used to solve the structure related to the Diels-Alder experiment is called single-crystal X-ray diffraction. It involves: 1) isolation of a single "perfect" crystal of specific size (0.1-0.3 mm edges) which is mounted on a thin glass fiber or capillary on a brass pin; 2) the capillary is then attached to an instrument called a goniometer which accurately rotates the crystal in 3-dimensions as an X-ray beam is directed at the crystal; 3) the periodic arrangement of atoms within the crystal diffracts the beam in a pattern observed by a film or detector; 4) the collected reflections (diffraction pattern) from all of the angles are processed by computer programs in order to determine the unit cell of the molecule (smallest volume unit containing all of the structural information that is repeated throughout the crystal lattice) and the relative 3-dimensional arrangement of the atoms in the structure. The resulting "solved structure" reveals the exact identity of each atom, its location, all bond distances, bond angles, and the overall molecular conformations. This information is invaluable especially when the chirality of a complex molecule cannot be determined by NMR, MS, or other spectroscopic tools. The CIF (Crystallographic Information File) or mmCIF (macromolecular CIF) can then be generated and used by programs such as ChemBio3D, Mercury, and Jmol to view the details of the solved structure in 3-dimension.

Assume that the X-ray crystal structure data given below corresponds to the product you isolated in lab.

X-ray Crystal Structure Data for the Major Product Isolated from the Diels–Alder reaction of Tetraphenylcyclopentadienone and Styrene

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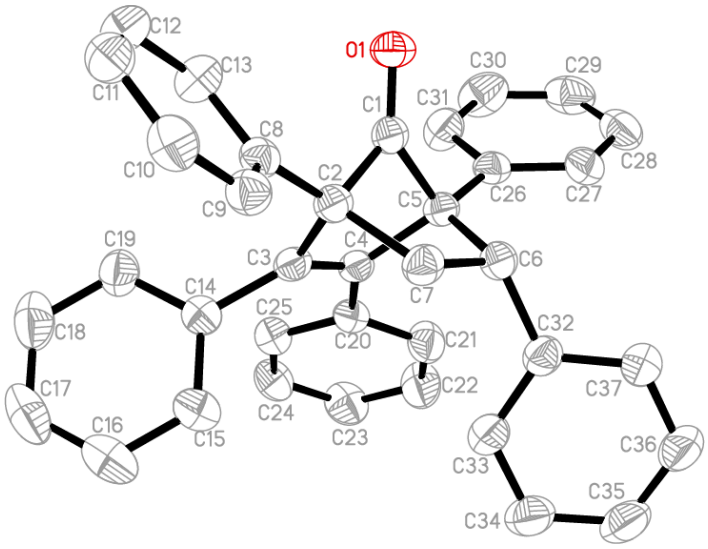


Figure 1: Thermal ellipsoid plot of Major Product showing 50% probability ellipsoids.

A crystal (approximate dimensions 0.42 x 0.33 x 0.25 mm^3) was placed onto the tip of a 0.1 mm diameter glass capillary and mounted on a CCD area detector diffractometer for a data collection at

173(2) K. [SMART V5.054, Bruker Analytical X-ray Systems, Madison, WI (2001).] From this analysis the illustration shown in Figure 1 – an ORTEP (Oak Ridge Thermal Ellipsoid Plot) depiction of the compound was generated. The shape and size of the thermal ellipsoids, in simple terms, represents the vibrational motion of each atom in a molecule and the probability of finding each atom's electron density within the ellipsoid sphere. For more information go to: http://www.crystallographiccourseware.com/ThermalParameters/ThermalParameters.html

The empirical formula was determined to be $C_{37}H_{28}O$, Formula Weight was calculated as 488.59.

Below is select data associated with the solved structure depicted in Figure 1.

Bond lengths (Angstroms)

O(1)-C(1)	1.2002(16)	C(17)-C(18)	1.384(3)
C(1)-C(2)	1.535(2)	C(17)-H(17A)	0.9500
C(1)-C(5)	1.5471(19)	C(18)-C(19)	1.383(2)
C(2)-C(8)	1.504(2)	C(19)-H(19A)	0.9500
C(2)-C(3)	1.5392(19)	C(20)-C(25)	1.395(2)
C(2)-C(7)	1.5670(19)	C(20)-C(21)	1.396(2)
C(3)-C(4)	1.356(2)	C(21)-C(22)	1.385(2)
C(3)-C(14)	1.484(2)	C(22)-C(23)	1.371(2)
C(7)-H(7A)	0.9900	C(6)-H(6A)	1.0000

Bond angles (degrees)

O(1)-C(1)-C(2)	131.47(13)	O(1)-C(1)-C(5)	129.52(13)
C(2)-C(1)-C(5)	99.00(11)	C(8)-C(2)-C(1)	117.69(12)
C(8)-C(2)-C(3)	118.03(12)	C(1)-C(2)-C(3)	96.18(11)
C(8)-C(2)-C(7)	117.29(12)	C(1)-C(2)-C(7)	98.36(11)
C(3)-C(2)-C(7)	105.67(11)	C(4)-C(3)-C(14)	129.22(13)
C(4)-C(3)-C(2)	109.10(12)	C(14)-C(3)-C(2)	121.35(12)
C(3)-C(4)-C(20)	128.39(13)	C(3)-C(4)-C(5)	108.34(12)
C(20)-C(4)-C(5)	123.14(12)	C(26)-C(5)-C(4)	117.65(12)
C(26)-C(5)-C(1)	113.28(11)	C(4)-C(5)-C(1)	96.64(10)
C(5)-C(6)-H(6A)	107.1	C(19)-C(18)-H(18A)	120.0

Torsion angles (degrees)

H6A-C6-C7-H7A =	- 12.60
H6A - C6 - C7 - H7B =	133.30